CS 473 (Spring 2025) Yiming Qin (yimingq3@illinois.edu)

Problem 6.2

Solution:

(a) For each triangle T_i, define

$$X_i := |\{p_{i,1}, \dots, p_{i,r}\} \setminus (T_1 \cup \cdots \cup T_{i-1})| \cdot \frac{\text{area}(T_i)}{r}.$$

$$a_i = \text{area}\left(T_i \setminus (T_1 \cup \cdots \cup T_{i-1})\right),$$

Consider each random point $p_{i,k}$ out of r points selected from T_i. Since the points are sampled uniformly, the probability that $p_{i,k}$ falls within the region $T_i \setminus (T_1 \cup \cdots \cup T_{i-1})$ is

$$\Pr\left[p_{i,k} \in T_i \setminus (T_1 \cup \cdots \cup T_{i-1})\right] = \frac{a_i}{\text{area}(T_i)}.$$

Each such point contributes $\frac{\text{area}(T_i)}{r}$, so its expected contribution is

$$\frac{\text{area}(T_i)}{r} \cdot \frac{a_i}{\text{area}(T_i)} = \frac{a_i}{r}.$$

Since there are r points sampled from T_i, the total expected contribution from T_i is

$$\mathbb{E}[X_i] = r \cdot \frac{a_i}{r} = a_i.$$

By linearity of expectation, sum over all X_i:

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} a_i.$$

The regions $T_i \setminus (T_1 \cup \cdots \cup T_{i-1})$ are disjoint and collectively form $T_1 \cup \cdots \cup T_n$, by defination,

$$\sum_{i=1}^{n} a_i = \text{area}(T_1 \cup \cdots \cup T_n) = A.$$

(b) Define

$$C_i = |\{p_{i,1}, \dots, p_{i,r}\} \setminus (T_1 \cup \cdots \cup T_{i-1})|$$
$$a_i = \text{area}\left(T_i \setminus (T_1 \cup \cdots \cup T_{i-1})\right),$$

we have,

$$C_i \sim \text{Bin}\left(r, p_i\right),$$

where

$$p_i = \frac{a_i}{\text{area}(T_i)}$$

is the probability that a random point in T_i is not covered by previous triangles.

$$\text{Var}[C_i] = r \cdot p_i \cdot (1 - p_i).$$

Thus,

$$\text{Var}[X_i] = \left(\frac{\text{area}(T_i)}{r}\right)^2 \cdot \text{Var}[C_i] = \frac{\text{area}(T_i)^2}{r^2} \cdot rp_i(1 - p_i) = \frac{a_i \text{area}(T_i)}{r}(1 - p_i).$$

Since $1 - p_i \leq 1$, we have:

$$\text{Var}[X_i] \leq \frac{a_i \, \text{area}(T_i)}{r}.$$

Since X_i are independent, the total variance is the sum of variances of X_i:

$$\text{Var}[X] = \sum_{i=1}^{n} \text{Var}[X_i] \leq \sum_{i=1}^{n} \frac{a_i \, \text{area}(T_i)}{r}.$$

Notice $\text{area}(T_i) \leq A$ for all i (since each triangle is part of the union). Thus,

$$\sum_{i=1}^{n} a_i \, \text{area}(T_i) \leq \sum_{i=1}^{n} a_i A = A \sum_{i=1}^{n} a_i = A \cdot A = A^2.$$

Substituting this bound:

$$\text{Var}[X] \leq \frac{A^2}{r}.$$

(c) For each triangle T_i, generating r random points takes $O(r)$ time. For each point $p_{i,j}$, checking membership in $T_1 \cup \cdots \cup T_{i-1}$ requires $O(i)$ operations (testing against $i - 1$ triangles, each in $O(1)$ time). Total time for all checks:

$$\sum_{i=1}^{n} r \cdot O(i) = O(rn^2).$$

Combining sampling and checks:

$$O(nr) + O(rn^2) = O(rn^2).$$

Choosing r as a constant gives $O(n^2)$ time.

By Chebyshev's Inequality, for $\epsilon = 0.01A$:

$$\Pr\left[|X - A| \geq 0.01A\right] \leq \frac{\text{Var}[X]}{(0.01A)^2} \leq \frac{A^2/r}{0.0001A^2} = \frac{10000}{r}.$$

To ensure $\frac{10000}{r} \leq 0.01$, set $r \geq 10^6$.